Benjamini-Yekutieli FDR

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Introduction

The Benjamini-Hochberg procedure controls FDR under independence or positive dependence. When p-values are arbitrarily dependent – including negatively correlated – BH's FDR guarantee can break. **Benjamini-Yekutieli (BY)** (2001) modifies BH by dividing by the harmonic number $c(m) = \sum_{i=1}^{m} 1/i$, giving FDR control under any dependence.

Prerequisites

Benjamini-Hochberg FDR, multiple testing.

Theory

Under arbitrary dependence, the BY procedure at level q:

1. Sort p-values: $p_{(1)} \leq \ldots \leq p_{(m)}$.
2. Find the largest k such that $p_{(k)} \leq kq/(m\,c(m))$ where $c(m) = \sum_{i=1}^{m} 1/i \approx \log m + 0.577$.
3. Reject all hypotheses with $p_{(i)} \leq p_{(k)}$.

The adjusted p-value is $\min(1, \min_{i \geq j} mc(m)p_{(i)}/i)$.

BY is more conservative than BH by roughly a factor of $\log m$: for $m = 1000$, $c(m) \approx 7.5$, so BY's threshold is about 7.5 times tighter.

In practice, BY is used mainly when:

- The p-values come from tests whose joint distribution is complex and cannot be shown to satisfy PRDS.
- The cost of a false discovery is high and independence cannot be asserted.

Assumptions

No dependence assumption – the procedure's FDR guarantee holds universally. The cost is reduced power.

R Implementation

```r
set.seed(2026)

# Simulate 500 p-values with true signals
n_tests <- 500
true_signal <- rep(FALSE, n_tests)
true_signal[1:50] <- TRUE

p_vals <- ifelse(true_signal,
                 pmin(runif(n_tests) * 0.02, 0.05),
                 runif(n_tests))

adj_bh <- p.adjust(p_vals, method = "BH")
adj_by <- p.adjust(p_vals, method = "BY")
```

```r
q <- 0.05
c(BH_rejections = sum(adj_bh < q),
  BY_rejections = sum(adj_by < q))

# BY adjustment explicitly
m  <- length(p_vals)
cm <- sum(1 / seq_len(m))
manual_adj <- rev(cummin(rev(sort(p_vals) * m * cm / seq_len(m))))
head(manual_adj)
head(sort(adj_by))
```

Output & Results

```
BH_rejections  BY_rejections
           47             24
```

BH identifies 47 signals at FDR 0.05; BY identifies only 24 at the same target. The additional conservatism of BY is visible as roughly halved power for this example.

Interpretation

Reporting: "Multiple testing was controlled at $q < 0.05$ using the Benjamini-Yekutieli procedure, which preserves FDR control under arbitrary dependence across tests."

Practical Tips

- Use BY when test dependence is complex and unknown.
- If tests are independent or positively dependent (common in RNA-seq), BH is strictly more powerful and equally valid.
- BY-adjusted p-values can be very close to 1 for large m; don't be surprised by unrecognisably inflated q-values.
- Simulation-based FDR control (via permutations) is often more powerful than BY when computationally feasible.
- The harmonic-factor multiplier makes BY nearly as conservative as Bonferroni for large m; consider whether you truly need arbitrary-dependence robustness.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests

- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests